Exercise 1: One-Hot Encoding for Product Categories
Dataset: Retail Dataset
This dataset contains transaction details from a grocery store, including product categories and purchase information.
Tasks

1. Load the dataset and select the columns Customer and Product_Category.
2. Apply **one-hot encoding** to the Product_Category column to prepare it for analysis, creating a binary column for each category.
3. Analyze the resulting dataset:
    - Discuss the impact of one-hot encoding on the data's size (number of columns).
    - Identify cases where one-hot encoding might lead to too many features and suggest alternative approaches.

Hint: use sklearn.preprocessing.OneHotEncoder or pandas.get_dummies()

Exercise 2: Label Encoding for Ordinal Data in Hotel Ratings
Dataset: Hotel Booking Demand
This dataset includes information about hotel bookings, including customer ratings and satisfaction levels.
Tasks

1. Select the Customer_Satisfaction column (or equivalent rating column) which ranks satisfaction levels from low to high.
2. Apply **label encoding** to transform the satisfaction levels into numerical values, ensuring the ordinal nature is preserved.
3. Evaluate the encoding results:
    - Discuss how this encoding works well for ordinal data and what might happen if label encoding was applied to non-ordinal data, like Hotel_Type (if available).

Hint: use sklearn.preprocessing.LabelEncoder

Exercise 3: Target Encoding for Cities in Used Car Prices
Dataset: Used Car Dataset
This dataset contains car listings, including features like City, Make, Model, and Price.
Tasks

1. Select the columns City and Price.
2. Apply **target encoding** on the City column by calculating the average car price for each city.
3. Analyze the target encoding results:
    - Discuss why target encoding can help reduce the number of features for high-cardinality data like cities.
    - Consider potential issues with data leakage if using target encoding in a supervised learning setting.

Exercise 4: Binary Encoding for State Codes in US Accidents Data
Dataset: US Accidents Data
This dataset includes data on accidents in the United States, with details like State, City, and Severity.
Tasks

1. Select the State column, which includes a large number of unique state abbreviations, and any other relevant columns like Severity.
2. Use **binary encoding** for the State column to keep the dimensionality manageable.
3. Reflect on the effectiveness of binary encoding:
    - Discuss why binary encoding is a good choice here compared to one-hot encoding, especially when dealing with many unique values.
    - Explain how binary encoding avoids imparting an arbitrary numeric order.

Hint: use category_encoders

Exercise 5: Frequency Encoding for Property Types in Airbnb Listings
Dataset: Airbnb Listings
This dataset includes Airbnb property listings with features like Property_Type, Room_Type, and Price.
Tasks

1. Choose the Property_Type column for encoding and analyze its unique values.
2. Apply **frequency encoding** by counting the occurrence of each property type and using this count as the encoded value.
3. Analyze the frequency encoding:
    - Discuss why this encoding can be useful for features with many unique values.
    - Describe any potential biases that might result from using frequency as an encoding method, especially if certain property types dominate the listings.